

4. 由 B 可知 A 有 3 个主元

$$\therefore \text{rank} A = 3$$

$$\therefore \text{rank} A + \dim \text{Null} A = 6$$

$$\therefore \dim \text{Null} A = 3$$

Col A 对应的基即为主元列

$\therefore$  Col A 的基为

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -1 \\ -3 \\ -2 \end{bmatrix}, \begin{bmatrix} 7 \\ 10 \\ -1 \\ -5 \\ 0 \end{bmatrix} \right\}$$

Row A 的基为 (写成列向量形式)

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ -3 \\ 7 \\ 9 \\ -9 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \\ 3 \\ 4 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \\ -1 \\ -2 \end{bmatrix} \right\}$$

$$\text{Null} A: \begin{cases} x_1 + x_2 - 3x_3 + 7x_4 + 9x_5 - 9x_6 = 0 \\ x_2 - x_3 + 3x_4 + 4x_5 - 3x_6 = 0 \\ x_4 - x_5 - 2x_6 = 0 \end{cases}$$

$$\begin{cases} x_1 = 2x_3 - 9x_5 - 2x_6 \\ x_2 = x_3 - 7x_5 - 3x_6 \\ x_4 = x_5 + 2x_6 \end{cases} \quad \therefore \text{Null} A \text{ 的基为 } \left\{ \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -9 \\ -7 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

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A为  $3 \times 4$  矩阵

$$\therefore \text{rank}(A) + \dim \text{Null} A = 4 \quad \text{rank} A \leq 4$$

$$\text{且 } \text{rank}(A) \leq 3$$

$\therefore \text{rank} A$  最大为 3

$\therefore A$  行空间最大维数为 3

A为  $4 \times 3$  矩阵

$$\text{rank}(A) + \dim \text{Null} A = 3 \quad \text{rank} A \leq 3$$

$$\text{rank} A \leq 4 \quad \therefore \text{rank} A \text{ 最大为 } 3$$

$\therefore A$  行空间最大维数为 3

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$$M = \begin{bmatrix} A & B \\ B & A \end{bmatrix}$$

M的行列式

$$\det(M) = \det(A+B) \times \det(A-B) \neq 0$$

$\therefore M$ 可逆

$$\text{令 } M^{-1} = \begin{bmatrix} C & D \\ E & F \end{bmatrix}$$

$$\begin{bmatrix} A & B \\ B & A \end{bmatrix} \begin{bmatrix} C & D \\ E & F \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

$$\text{即 } \begin{cases} AC + BE = I \\ AD + BF = 0 \\ BC + AE = 0 \\ BD + AF = I \end{cases}$$

解方程组可知

$$M^{-1} = \frac{1}{2} \begin{bmatrix} (A+B)^{-1} + (A-B)^{-1} & (A+B)^{-1} - (A-B)^{-1} \\ (A+B)^{-1} - (A-B)^{-1} & (A+B)^{-1} + (A-B)^{-1} \end{bmatrix}$$

$$2. A^2 - 2A - 3I = 0$$

$$(A - 3I)(A + I) = 0$$

$$\det(A - 3I) \det(A + I) = 0$$

$$\text{则 } \det(A - 3I) = 0 \text{ 或 } \det(A + I) = 0$$

没法证明  $A + I$  可逆

3. ①

$$(A+B)(A+B)$$

$$= (A+B)A + (A+B)B$$

$$= A^2 + AB + BA + B^2$$

$$= A+B+AB+BA = A+B$$

$$\therefore AB+BA=0$$

又

$$AB = AAB = -ABA$$

$$= BAA = BA$$

$$\therefore AB = BA = 0$$

0.3



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4. 由矩阵乘法可知  $AB \in R^{m \times p}$

证明  $\text{rank}(AB) \leq \text{rank}(A)$

$AB = C$  则  $C$  的列向量  $C_i = A(B_i)$

$B_i$  为  $B$  的列向量.

说明  $C$  的列向量是  $A$  的列向量组的线性组合

$\therefore \text{rank}(AB) \leq \text{rank}(A)$

对  $AB$  进行转置  $(AB)^T = B^T A^T$

由前面证明可知  $\text{rank}(B^T A^T) \leq \text{rank}(B^T)$

又  $\therefore \text{rank}(AB) = \text{rank}(AB)^T = \text{rank}(B^T A^T)$

$\text{rank}(B) = \text{rank}(B^T) \therefore \text{rank}(AB) \leq \text{rank}(B)$

$\therefore \text{rank}(AB) \leq \min \{ \text{rank}(A), \text{rank}(B) \}$

5.  $\therefore A^2 = A \therefore A(A-I) = 0 \quad (A-I)A = 0$

$\therefore AB = 0 \therefore \text{rank}(A) + \text{rank}(B) \leq n$

证  $\therefore AB = 0 \therefore \dim \text{Null } A \geq \text{rank } B$

这是因为  $\text{Null } A$  是所有形如  $Ax = 0$  的  $x$  组成的空间而

$B$  的列空间全部向量都满足,  $\therefore \text{rank}(A) + \dim \text{Null } A = n$

$\therefore \text{rank}(A) + \text{rank}(B) \leq n$

这里  $B = A - I$

$$\therefore \text{rank}(A) + \text{rank}(A - I) \leq n$$

$$\text{又} \because n = \text{rank}(I) = \text{rank}(A + (I - A))$$

$$\leq \text{rank}(A) + \text{rank}(I - A) = \text{rank}(A) + \text{rank}(A - I)$$

$$\therefore \text{综上} \text{rank}(A) + \text{rank}(A - I) = n$$

0.3



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$$5.1 \quad 5 \quad 16 \quad 21$$

$$5. \quad A = \begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 3-\lambda & 7 & 9 \\ -4 & -5-\lambda & 1 \\ 2 & 4 & 4-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) =$$

$$(3-\lambda) \begin{vmatrix} -5-\lambda & 1 \\ 4 & 4-\lambda \end{vmatrix} + (-7) \begin{vmatrix} -4 & 1 \\ 2 & 4-\lambda \end{vmatrix}$$

$$+ 9 \begin{vmatrix} -4 & -5-\lambda \\ 2 & 4 \end{vmatrix}$$

$$= -(3-\lambda)((5+\lambda)(4-\lambda) - 4) - 7(4(\lambda-4) - 2) + 9(-16 + 2(5+\lambda)) = -\lambda^3 + 2\lambda^2 + 17\lambda$$

$$\text{令 } \det(A - \lambda I) = 0$$

$$\text{得 } \lambda_1 = 0 \quad \lambda_2 = 1 + 3\sqrt{2} \quad \lambda_3 = 1 - 3\sqrt{2}$$

∴特征值为  $\lambda = 0$  时

$$(A - \lambda I)x = 0$$

$Ax = 0$  对  $A$  进行行变换化简后求出

$$x = \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix} \text{ 为特征向量.}$$

用同样的方法,  $\lambda = 1 + 3\sqrt{2}$  时

$$x = \begin{bmatrix} \frac{3\sqrt{2} + 23}{14} \\ \frac{9\sqrt{2} - 22}{14} \\ 1 \end{bmatrix}$$

$$\lambda = 1 - 3\sqrt{2} \text{ 时 } x =$$

$$\begin{bmatrix} \frac{-3\sqrt{2} + 23}{14} \\ \frac{-9\sqrt{2} - 22}{14} \\ 1 \end{bmatrix}$$



16.

$$A = \begin{bmatrix} 3 & 0 & 2 & 0 \\ 1 & 3 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} \quad \lambda = 4$$

$$A - 4I = \begin{bmatrix} -1 & 0 & 2 & 0 \\ 1 & -1 & 1 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

行化简为阶梯形矩阵:

$$\begin{bmatrix} -1 & 0 & 2 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = 2x_3$$

$$x_2 = 3x_3$$

则通解为

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

一个基为

$$\left\{ \begin{bmatrix} 2 \\ 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

21 a.  $X$  必须有非平凡解.

b.  $\checkmark$

c.  $\checkmark$

d.  $\checkmark$

e.  $X$  行化简可以求特征向量  
但不能求特征值